

First Z77 Ivy Bridge boards

Manufacturers have started revealing motherboards based on the Z77 chipset for Intel's Ivy Bridge processors at CES 2012

Diablo III in the UK

By the time you're reading this, Diablo III will have released on 3rd February in the UK. Look out for the review in coming issues

The future is 3D

Autodesk is a company that needs no introduction. Some may say it's the world leader in 3D and engineering design software. We recently met with Rob Hoffman, Senior Product Marketing Manager, 3D Media and entertainment at Autodesk Inc., and discussed the future of Autodesk. Here's what he had to say

The next big thing

There's so much going on in the 3D industry. Movies such as *Avatar* have spoiled the audience with special effects and animation and their expectations have only gone up. We want more complexity, more detail, the next movie needs to further blow our mind. Designers are being asked to work on bigger and more complex projects, but with less time and a smaller budget. What Autodesk is trying to do is bridge that gap between expectations and resources. We're trying to do that with what we're calling "Digital Entertainment Creation", which is basically to make the creation of content easier for everyone, while still giving them the power to be completely creative. To do this, we're working primarily on standardisation.

The need for standardisation and distributed production

When it comes to creating something like a movie, there are a lot of tool sets involved. These tool sets need to be able to "talk" to one another. It should be easy for artists to switch between such multiple tool sets and get the work done. Using just one software such as Maya or 3DS Max is not necessarily the most efficient way to go about creating your work; you'll need multiple tools and it's essential that they



work together. One way Autodesk wants to do this is by standardising the formats and even the interface elements, allowing artists to switch between their tool sets and not feel lost. Autodesk also wants to work on providing a broader palette of tools for artists to draw from.

Autodesk has also adopted a format, which was developed by Sony Imageworks and Lucasfilms, known as Alembic. The main advantage of which is the comparatively small file size allowing for better portability. This has the added advantage that artists and production studios all over the world can work on the same project 24x7, without worrying about compatibility and bandwidth issues.

Virtual Production and Avatar!

Now this is a very interesting concept. As you know, movies and games, in fact, a scene of any sort, starts off as a storyboard. The director works with an artist and sketches out a scene in the same way you would create a comic strip. This allows the director to visualise, in a very rudimentary fashion, what the scene would look like. Autodesk has taken this to the next level using Virtual Production.

You can now generate a virtual storyboard and even view a CG scene in real-time, while the characters are enacting it. James Cameron did this in *Avatar*, by setting up his tablet as a virtual camera. His tablet was tracked by the software (using motion capture), and its position and orientation was interpreted as a camera within the software. At the same time, the 3D scene being rendered was streamed to the tablet as a video. This allowed him to walk around the set and observe elements as they would be seen in the movie (the blue screen would be replaced by the actual environment of the scene), enabling him to direct the actors in real-time in the context of the animated environment.

Autodesk recently signed an agreement with Walt Disney for a tech called "X-gen", developed by Walt Disney's top secret development division. The technology allows artists to procedurally insert various items and effects into a scene. A good example would be *Tangled*, where the character's hair was rendered by this method. Even the *Pirates of the Caribbean*, where Davy Jones's tentacles underwent the same treatment.

Social/Casual gaming and Autodesk

Everybody plays games now. The gaming demographic has gone from the 14-19 years of yore to the current small kid-to-grandma stage. Autodesk realises that and has decided to take a very close look at social and casual games. While it really doesn't have a suite or application specifically designed for this genre, it's working on it and hope to have one ready in the near future.

Parting thoughts

Autodesk has always been looked on as the company that made the software for developing things like in-game assets and effects such as weapon models and muzzle flashes. Now however, Autodesk wants to improve on the middle-ware it provides and has been working on various pieces of software such as Kynapse and HumanIK as an AI and Inverse Kinematics system for use in animation and games. **d**

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